

Explorations in Computer Vision

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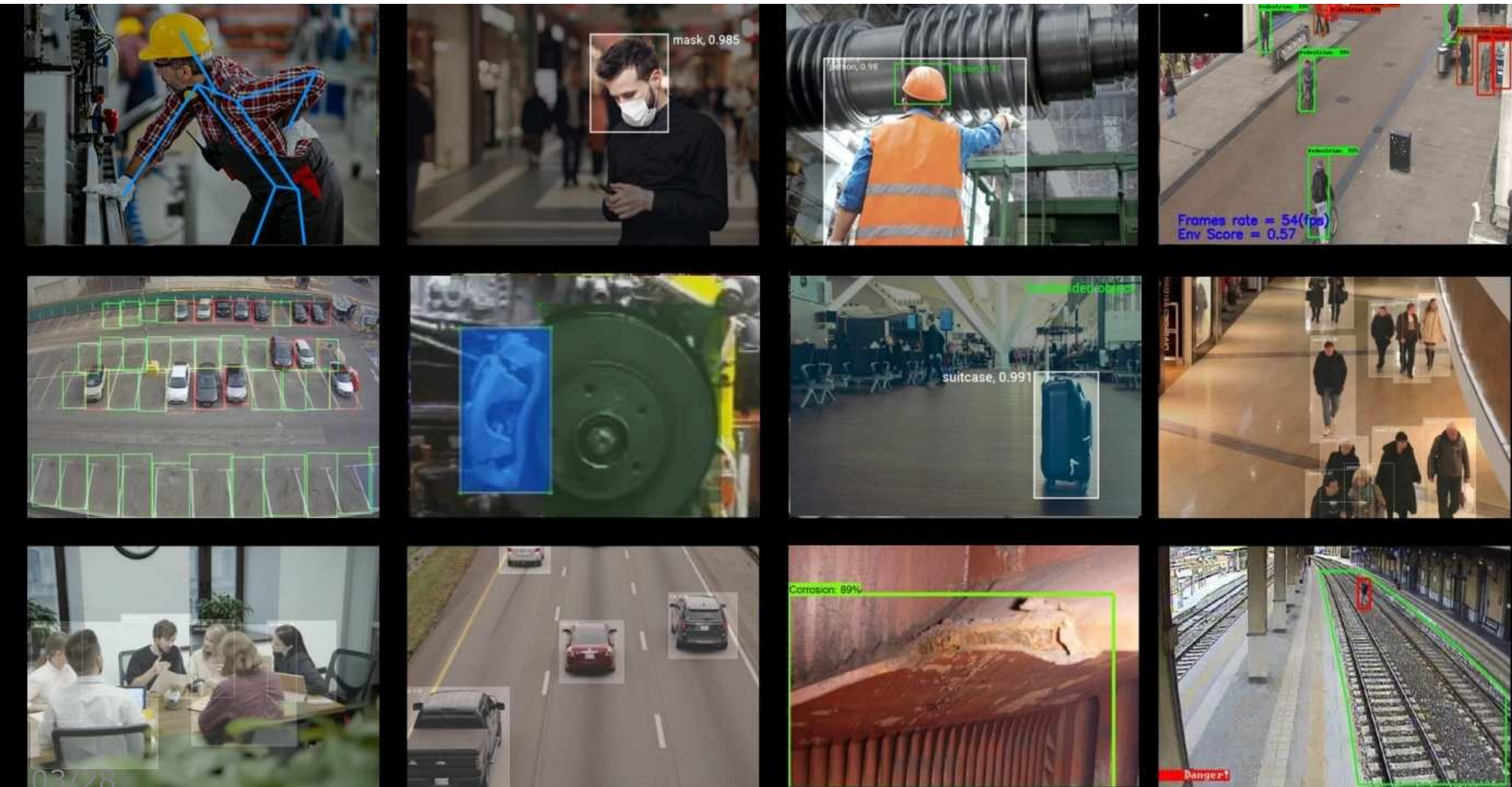
Overview

Computer vision systems have many applications
Very powerful to be able to detect our environment

Custom sensors can be expensive and inflexible
Cameras and analysis software much more versatile

Let's look at some example applications...

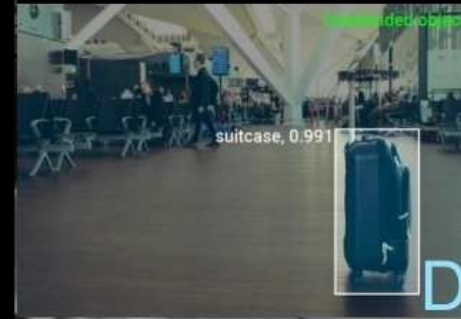
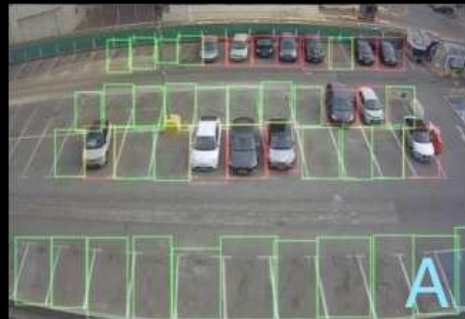
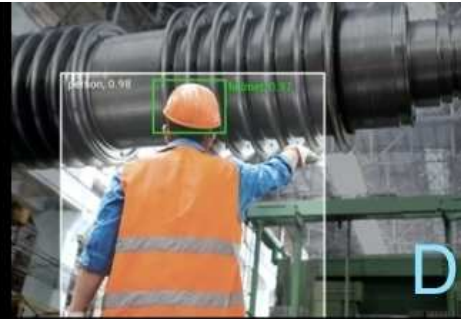
Example Computer Vision Applications



Different Levels of Comprehension

- Attention: is there *something* (anything) present ?
- Detection: specific *type* of thing (car, person etc) ?
- Recognition: which *individual* thing (car, face etc) ?
- Interpretation: what is the thing actually *doing* ?

Classification



Our Approach

We **could** try to train a machine learning model
In order to identify objects/structures in an image
But this wouldn't give much insight into the process
(We'd just be training, rather than experimenting)

Instead we will take a much lower-level approach
Focusing on writing code to analyse images/video
Using a number of pixel manipulation techniques...

Pixel Manipulation Techniques

- Searching for specific colours (RGB, Hue, Sat. etc.)
 - Frame differencing (change in Brightness, Hue etc.)
 - Locations sensing (position of artefacts is important)
 - Movement detection (change in location is useful)
-
- Scanning for specific shapes (person, car, letter etc.)
 - Matching relative structures (e.g. face detection)

The Application

We need some form of visual material to process
For this, we will use a recording of a webcam stream
From the "Marine Biological Association" in Plymouth:

[3m5!1s0x486c935311b11e0b:0xd40b5a4f7f597d10!8m2!3](#)

It's a nice view, with lots of activity going on !

video

Your Task

Shown in the camera stream is a busy road junction

Your task: analyse traffic passing through this junction

Don't panic: step-by-step instructions will be provided !



Processing Template

To get you started, we've provided a template to:

- Load in an MP4 video file
- Extract a single image ("frame") from the video
- Draw that frame onto the screen
- "Mask off" unwanted areas of the frame
- Define a colour range for analysis
- Analyse remaining "unmasked" pixels within frame

VehicleAnalysisTemplate

Drawing Pixels onto a Frame

To mask off areas of the frame:

- Set the "fill" colour to black (0,0,0)
- Then draw Rectangles and Triangles

To draw individual pixels:

- Set "stroke" colour - for example red (255,0,0)
- Then draw individual "points" (pixels) in that colour

Examples of both the above are in the template code

Checking the Colour of Individual Pixels

We can get the colour of a particular pixel using:

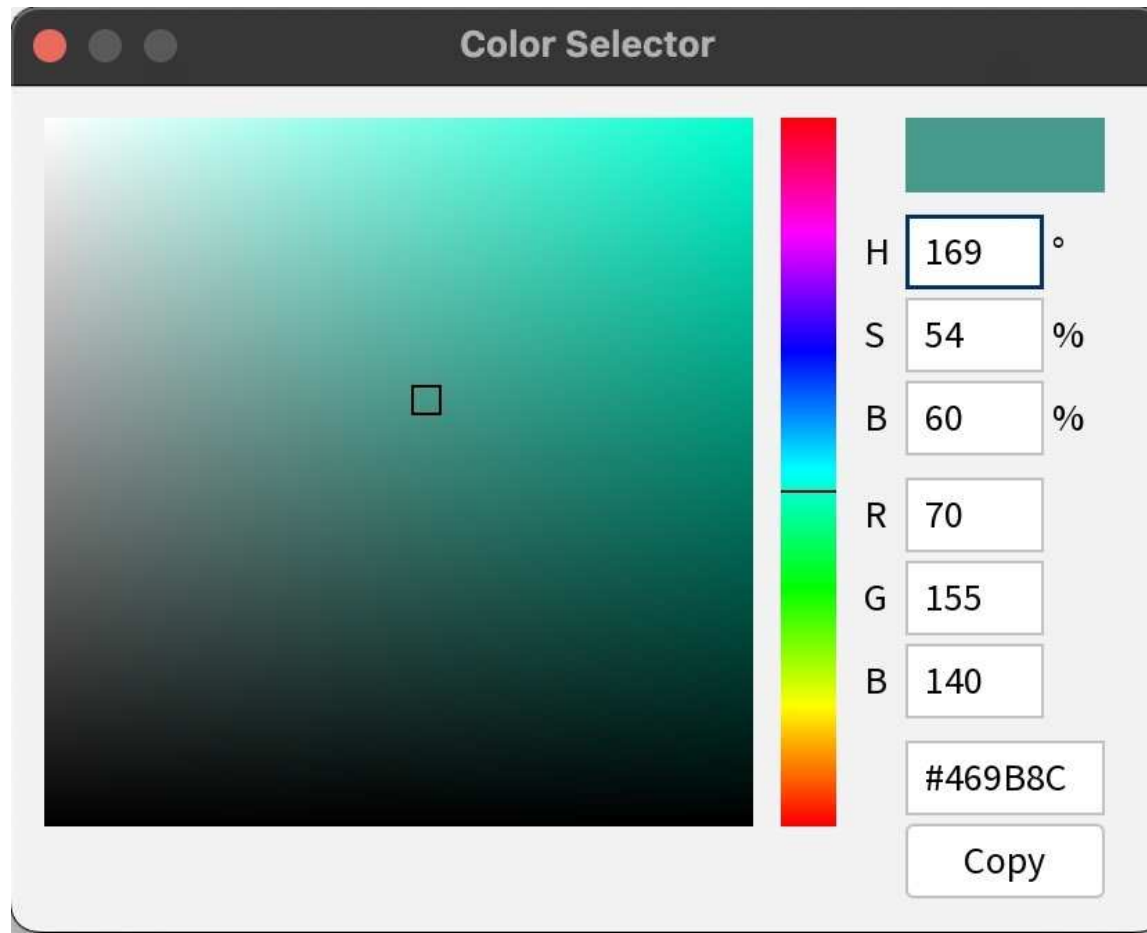
```
int pixelColour = currentFrame.get(x,y);
```

Then extract various properties of the pixel using:

```
int redness = red(pixelColour);  
int greenness = green(pixelColour);  
int blueness = blue(pixelColour);  
int whichColour = hue(pixelColour);  
int howBright = brightness(pixelColour);  
int howSaturated = saturation(pixelColour);
```

Hue / Saturation / Brightness / RGB

Warning: All values in Processing are in range 0-255



IMPORTANT NOTE

This is a very processor-intensive activity !
280k pixels per frame requires a LOT of calculation

If you do not have a fast laptop
We advise you to use a lab machine !!!
(otherwise it will be painfully slow !)

Instructions

Go to the summer school GitHub page:
<https://github.com/UoB-SummerSchool/>

Clone the repository for this exercise:
CameraAnalysisAndComms

Work through the tasks in the instructions document

Libraries

In order to run the template project
You will need to install a couple of libraries:

Sketch > Import Library > Manage Libraries

The screenshot shows the 'Libraries' panel in the Arduino IDE. It has tabs for 'Libraries', 'Modes', 'Tools', 'Examples', and 'Updates'. The 'Libraries' tab is active, displaying two lists of installed libraries. Each list has a search bar at the top and a dropdown menu set to 'All'. The first list shows the 'MQTT' library by 'Joel Gaehwiler'. The second list shows the 'Video Library for Processing 4' by 'The Processing Foundation'.

Status	Name ↓	Author
	MQTT	
	MQTT MQTT library for Processing based on the Eclipse Paho pr...	Joel Gaehwiler

Status	Name ↓	Author
	Video 4	
	Video Library for Processing 4 GStreamer-based video library for ...	The Processing Foundation

To Work !

Remote Notification

Remote Notifications

It's no good being able to detect something...

If nobody actually realises the event has occurred !

We might even want to notify multiple remote users
(In case many people are interested in the event)

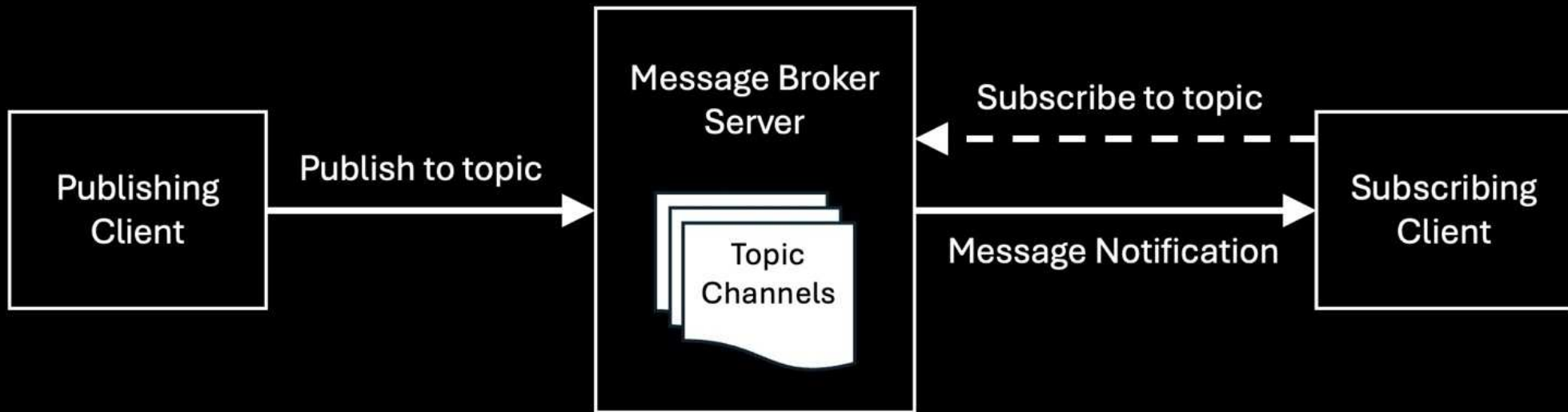
There are various mechanisms available to do this
We use a communication mechanism called MQTT...

Message Queue Telemetry Transport (MQTT)

Message passing protocol (ideal for notifications !)

You can "publish" messages to "topics" (channels)

Other people "subscribe" to receive notifications



Message Queue Telemetry Transport (MQTT)

Connect to broker by adding this to setup method:

```
mqtt = new MQTTClient(this);  
mqtt.connect("mqtt://broker.hivemq.com:1883");
```

To publish a notification message to a topic we use:

```
mqtt.publish(topic, parameter);
```

Topic should start with "junction/" for example:

```
mqtt.publish("junction/vehicles/car/red", "North");
```

Alternative Servers

A server that gathers messages is called a "Broker"
(Arbitrates between publishers and subscribers)

For this activity we are using a free public broker
Since it's free there are no guarantees of availability
Sometimes the server is not operational :o(

That's fine - there are other alternatives available:

`mqtt://broker.emqx.io:1883`

`mqtt://broker.hivemq.com:1883`

`mqtt://test.mosquitto.org:1883`

`mqtt://public-mqtt-broker.bevywise.com:1883`

Central Projection Screen

We'll run a message monitor on the projector screen:

`MultiNotificationMonitor`

"plymouth" messages published to brokers will appear

Let's just run a quick test to make sure it is working:

`NotificationTester`

To Work !

Bonus Slides

What is happening right now ?

<http://webcam.mba.ac.uk/mjpg/video.mjpg>

Want test your code analysing the live video feed ?

Find all instances of: `drawMovieOntoFrame`

And replace them with: `drawLiveStreamOntoFrame`

For example, the following line:

```
drawMovieOntoFrame(currentFrame);
```

Would now become:

```
drawLiveStreamOntoFrame(currentFrame);
```

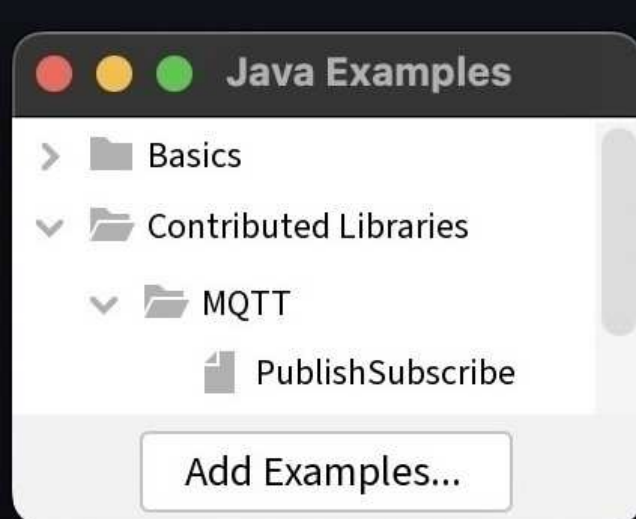
Be careful - try not to behave like a DoS attack !

What else is going on ?

Do you want to see what everyone else is doing ?

You can create a new Processing project to find out !

Take a look at the MQTT examples (File>Examples...)



```
void clientConnected() {  
    println("client connected");  
    client.subscribe("/hello");  
}  
  
void messageReceived(String topic, byte[] payload) {  
    println("message: " + topic + " - " + new String(payload));  
}  
  
void connectionLost() {  
    println("connection lost");  
}
```

Blob Detection

Counting pixels is a simple but effective approach

Blob Detection is a more advanced technique

Involves scanning for coherent clusters of pixels

(Based on the pixel brightness or colour)

Narrowing down the bounding box to get a tight fit

BlobDetection